



Process Cooling

Flexible Solutions for Liquid Cooling

Well matching components · Optimized systems and services ·
Flexible configurations with a unified concept of operation



Protecting man, machine and the environment.

WE MAKE COMPANIES MORE PRODUCTIVE.

For more than 60 years, the Pfannenberg Group has been helping to guarantee production safety for companies throughout the world. Our mission is to satisfy the increasing demands of modern industries by developing progressive ideas. This led to the invention of many products in the field of thermal management, signaling technology and process cooling.

A spirit of invention and German engineering genius are not our only strengths. We are also proud of the close relationships we maintain with our clients and their industries.

Pfannenberg's experience in delivering a broad range of cooling solutions has led to a wide product range and a team of experts that shares its competencies with customers worldwide. We develop customised high quality, cost effective, energy efficient solutions for demanding industrial requirements. This is the real value for our customers.

To understand more about how we can help you, please contact your Pfannenberg local sales office. As one of the few companies around the world with

a complete range of industrial thermal management solutions and services, we have a wide range of expertise which we would like to share with you.

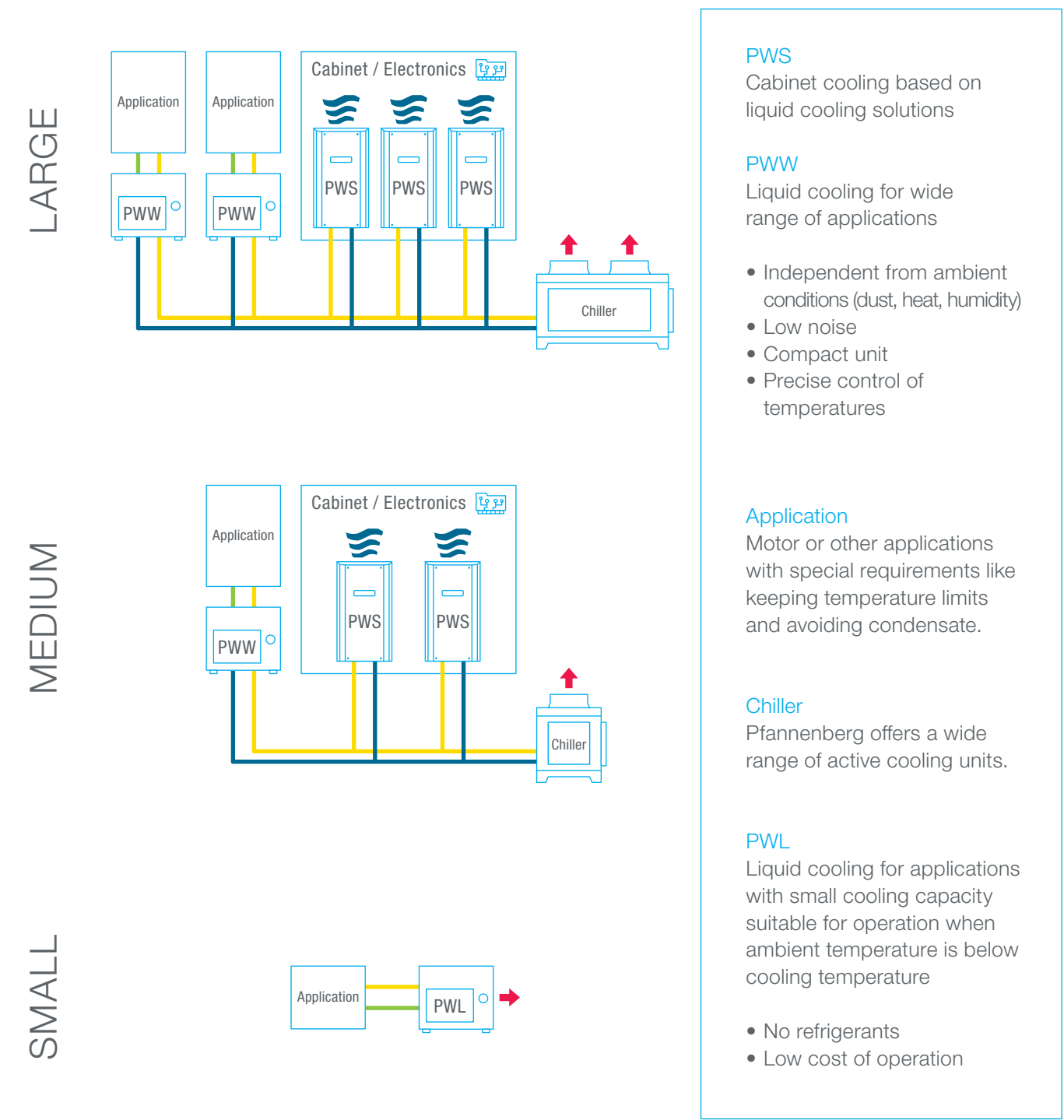
Being a global team, we place the knowledge and technical expertise of our engineers and experts at your disposal so that you can find the best possible solutions for your requirements. Today and in the future.

What can we do for you?

Marcello Chiozza
Global Product Manager
Liquid Cooling Solutions

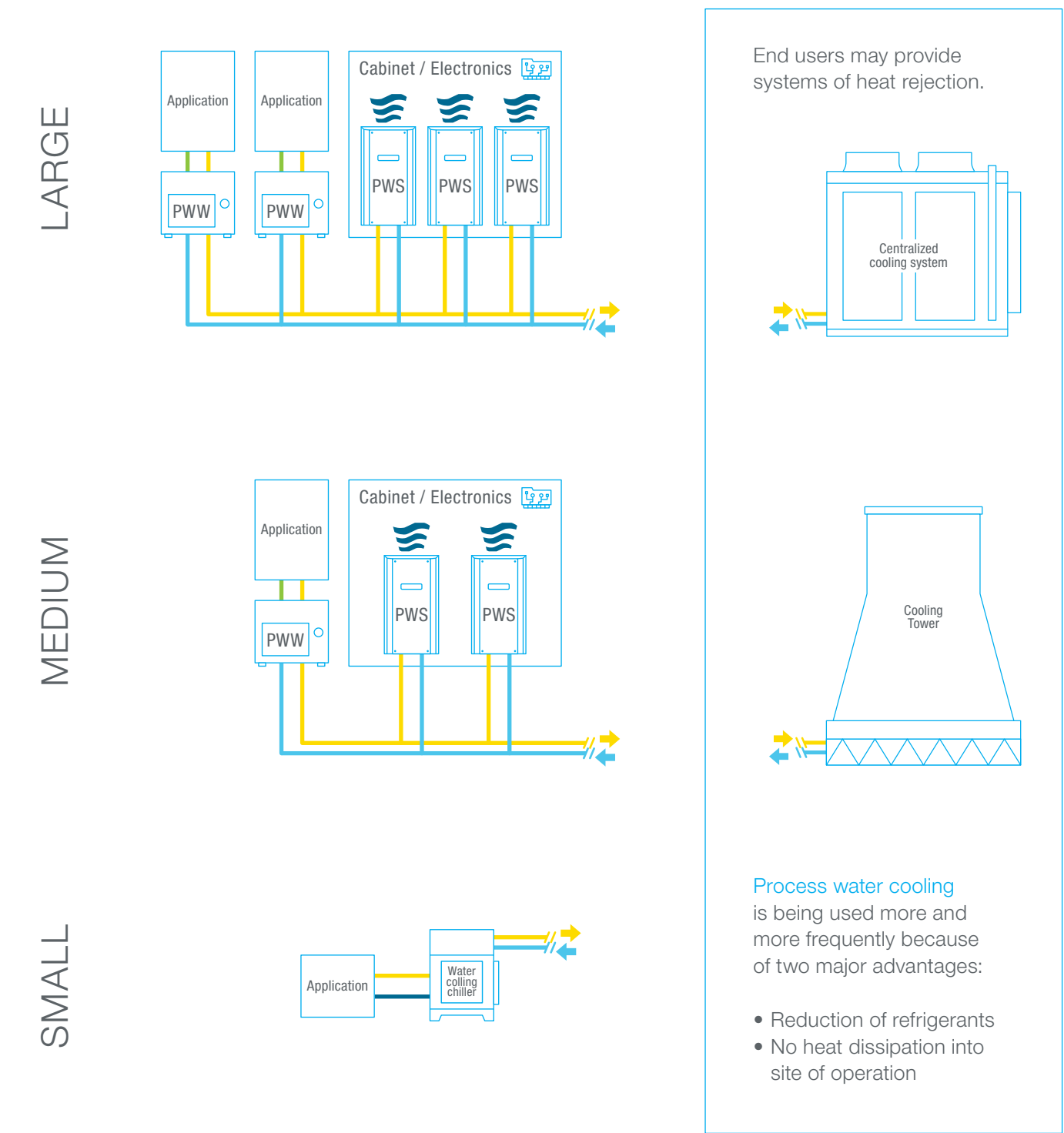


Pfannenberg liquid cooling solutions for decentralized cooling applications



- Hot air from chiller (active or passive unit) into site of operation
- Cold water provided by active cooling unit – low temperature due to required cooling capacity
- Cool water provided by passive cooling unit (PWW/PWL) – temperature matching requirements of application

Pfannenberg liquid cooling solutions for centralized cooling applications



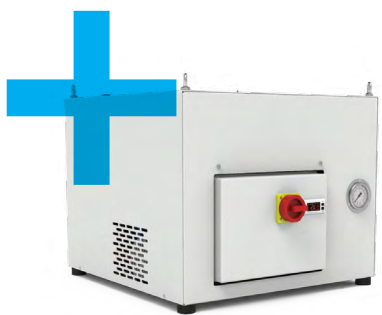
- Cold air provided by passive cooling unit (PWS)
- Process water from local source (process water)
- Warm water returning from application – temperature depending on actual heat load

Reflecting many years of experience in numerous areas of the industry: State-of-the-art standard units

The plus to protect against condensate and corrosion. Optimizing energy consumption and costs.



- PWS**
- Air / water heat exchangers
 - Mounted on door or side
 - 0,5 – 10 kW
 - + 1°C / +70°C *



- PWW**
- Water / water heat exchanger
 - Stand above or for interstices
 - 3 – 40 kW
 - +10°C / +50°C *



- PWL**
- Water / Air heat exchanger
 - Stand above or for interstices
 - 1 – 10 kW
 - +5°C / +40°C *

* Range of ambient air or recooling water temperatures suitable for operation of the units.



UL approval



service friendly



inverted technology available



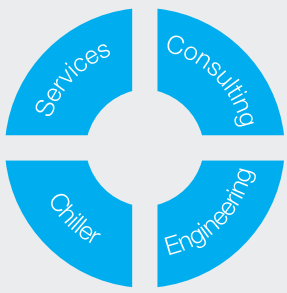
microchannel technology



protection system



extreme environmental condition



Visit us online.

Online you can get comprehensive information on all our liquid cooling devices. Additionally, you are able to download product data sheets, manuals, declarations of conformity and certificates.

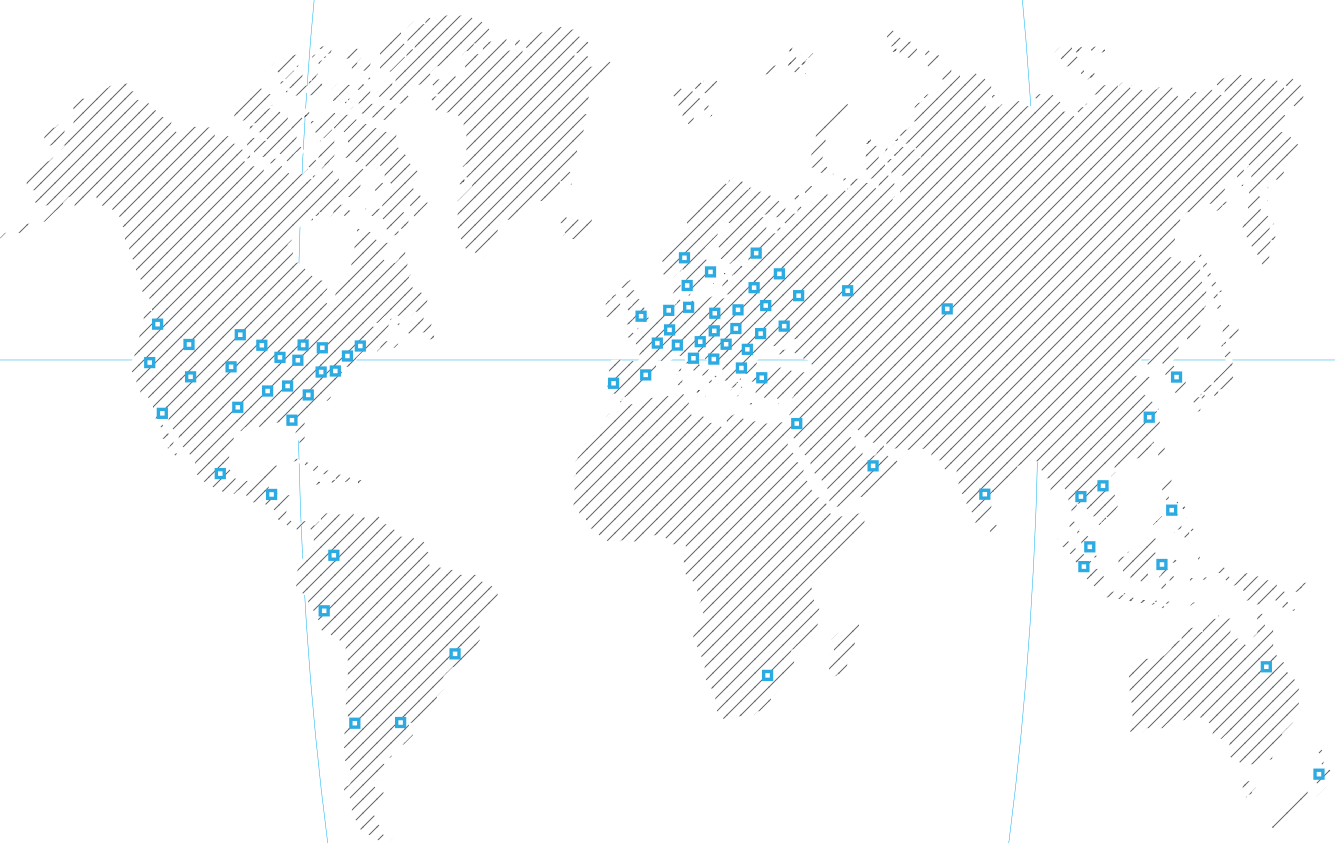
PFANNENBERG.COM/LIQUIDCOOLING



FROM PLANNING TO SERVICES

Pfannenberg offers the full range of products and support to help you creating and operating reliable and efficient cooling solutions with only one contact for all questions.

ALL AROUND THE WORLD.



CUSTOMIZED LIQUID COOLING

Pfannenberg passive cooling units are key elements of high performance and low maintenance cooling solutions matching a wide range of industrial applications. They can be adjusted to the application and provided as customer-specific equipment.

CUSTOM SOLUTIONS.

The Pfannenberg group worldwide

Pfannenberg Europe GmbH
Werner-Witt-Straße 1
21035 Hamburg
Germany

Phone: +49 40 73412 156
Telefax: +49 40 73412 101
Email: info@pfannenberg.com
Web: www.pfannenberg.com

Pfannenberg Austria, Ottlang am Hausruck
Phone: +43 7676 50219
Email: info.austria@pfannenberg.com

Pfannenberg Brazil, Indaiatuba
Phone: +55 19 3935 7187
Email: info@pfannenberg.com.br

Pfannenberg China, Suzhou
Phone: +86 512 6287 1078
Email: info@pfannenberg.cn

Pfannenberg France, Rueil-Malmaison
Phone: +33 1 4708 4747
Email: info@pfannenberg.fr
Pfannenberg Italy, Fidenza (PR)
Phone: +39 0524 516 711
Email: info@pfannenberg.it

Pfannenberg Poland, Warsaw
Phone: +48 228907246
Email: info@pfannenberg.pl

Pfannenberg Russia, St. Petersburg
Phone: +7 812 612 8106
Email: info@pfannenberg.ru
Pfannenberg Singapore, Singapore
Phone: +65 6293 9040
Email: info@pfannenberg.com.sg

Pfannenberg United Kingdom, Rotherham
Phone: +44 1709 36 4844
Email: info@pfannenberg.co.uk

Pfannenberg USA, N.Y.
Phone: +1 716 685 6866
Email: info@pfannenbergusa.com

Deliveries are made on the basis of the General Terms and Services of the ZVEI.
Subject to technical amendments and misprints. This paper has been manufactured
from chlorine-free bleached cellulose.



075000295